



SIGNAL ANALYSIS REPORT

Automated Signal Analysis and Parameter Extraction

SIH Signal Analyzer • National Technical Research Organisation

Date / Time: 2026-09-29 22:53:07

Input File: synthetic_BPSK_1000kHz_SNR20dB.wav

Format: WAV Audio / IQ (.WAV)

1. FILE / SIGNAL INFORMATION

File Name	synthetic_BPSK_1000kHz_SNR20dB.wav
File Format	WAV Audio / IQ (.WAV)
File Size	69,676 bytes
Signal Duration	17.41 ms (0.01741 s)
Sample Rate (Fs)	1.0000 MHz (1,000,000.0 Hz)
Center Freq Offset	-4.88 Hz
Total Samples	17,408

2. DETECTED SIGNAL PARAMETERS

Modulation	BPSK
Modulation Confidence	100.00%
Occupied Bandwidth	173.828 kHz (173,828.1 Hz)
Carrier Freq Offset (CFO)	-4.88 Hz
Symbol Rate	125,000.0 symbols/sec
Samples Per Symbol	8 sps
Frame Sync Status	Locked

3. MODULATION CLASSIFICATION

Modulation Scheme	Probability
BPSK	100.00%
QPSK	0.00%
GMSK	0.00%

4. TIME-DOMAIN WAVEFORM

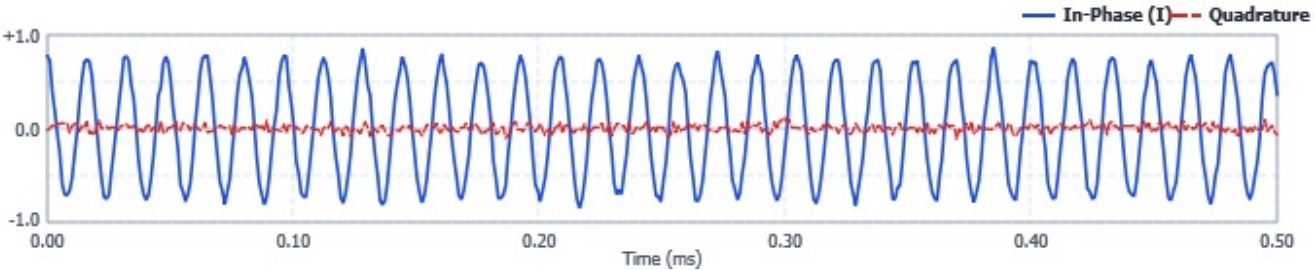


Figure 1: Time-Domain Waveform (In-Phase I & Quadrature Q Components)

5. FREQUENCY SPECTRUM

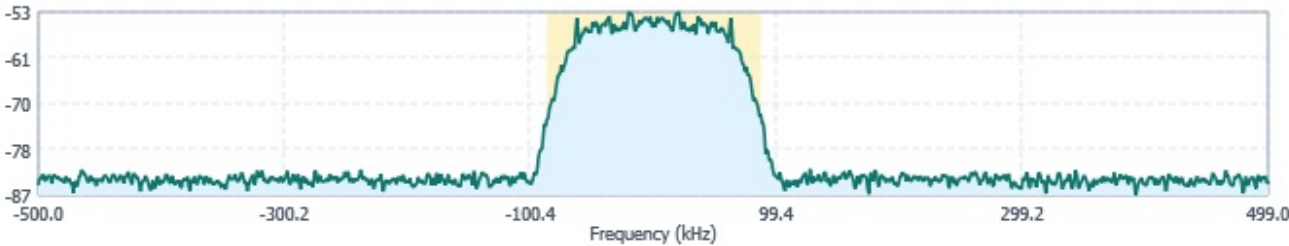


Figure 2: Power Spectral Density (PSD) and Occupied Bandwidth

6. SPECTROGRAM / WATERFALL

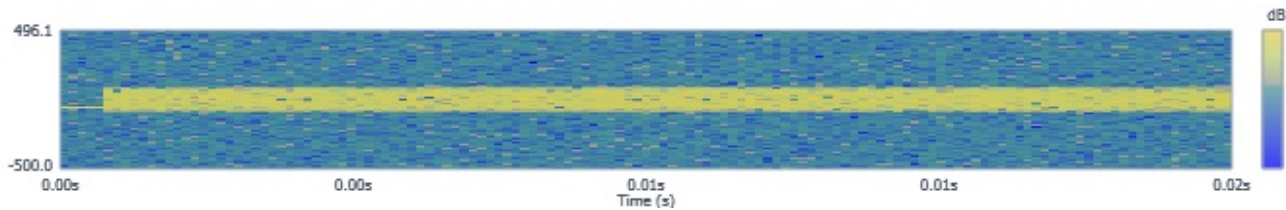
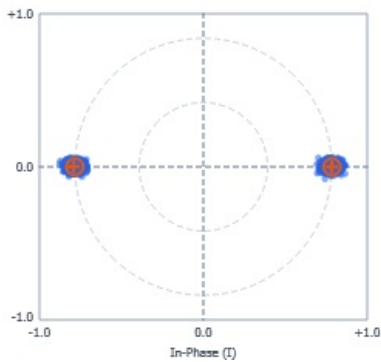


Figure 3: Time-Frequency STFT Spectrogram Waterfall

7. CONSTELLATION DIAGRAM



8. SYNCHRONIZATION / FRAME ANALYSIS

Frame Sync Status	Locked
Header Offset	110 bits
Payload Offset	142 bits
Sync Pattern Locked	CCSDS_ASM
Correlation Sync Match	100.0%
Phase Ambiguity Resolved	Yes (Costas ambiguity resolved)
Demodulated Bits	2,083 bits

Figure 4: IQ Constellation

9. BITSTREAM / DECODING

Total Decoded Bits	1,941 bits	Decoded Bytes	243 bytes
Bit Density (1s / 0s)	954 (49.1%) / 987 (50.9%)	ASCII Text Preview	..+....}Rp..S.>+..g...0.~@..[./.: \$.6d.rI.j5.>...Ru../T....~... 3..Vnrb.b..
Bitstream Sample	10101101 11110000 00101011 11000111 11101000 10111011 00010011 01111101 01010010 01110000 00011110 11100001 ... (+1845 bits)		

Decoded Bytes Hex Dump:

000000	AD F0 2B C7 E8 BB 13 7D 52 70 1E E1 53 C7 3E 00	..+....}Rp..S.>.
000010	2B D8 7F 67 89 1A 85 4F B1 7E 40 04 F4 5B DF 2F	+..g...0.~@..[./
... (211 additional bytes omitted)		

10. FEC / DE-INTERLEAVER ANALYSIS

Component	Detected Configuration	Status / Confidence
De-interleaver	none	Bypassed
Forward Error Correction (FEC)	none	Uncoded
Channel Decoder	Automated Multi-Scheme Pipeline	243 bytes recovered

11. FINAL ANALYSIS SUMMARY

RF capture file 'synthetic_BPSK_1000kHz_SNR20dB.wav' was processed, containing 17,408 samples recorded at a sample rate of 1.0000 MHz (1,000,000.0 Hz) (effective duration: 17.41 ms (0.01741 s)). The neural modulation classifier identified the transmission format as BPSK with a classification confidence of 100.00%. Spectral analysis determined an occupied bandwidth of 173.828 kHz (173,828.1 Hz) with a carrier frequency offset of -4.88 Hz. Demodulation recovered 2,083 raw bit transitions. Frame synchronization locked to CCSDS_ASM (100% match) with header boundary at bit 110 and payload commencing at bit 142. Channel decoding detected direct (no) interleaving, and raw uncoded bit framing, successfully assembling 243 decoded payload bytes.